PRODUCT USER MANUAL

For Global Ocean Multi Observation OMEGA3D product

MULTIOBS_GLO_PHY_UVW_3D_MYNRT_015_007
Issue: 3.0

Contributors : B. Buongiorno Nardelli, S. Asdar, L. Della Cioppa

Approval Date: November2025

MERCATOR OCEAN
INTERNATIONAL
2 avenue de l'Aérodrome de Montaudran, 31400 Toulouse, FRANCE
Tél : +33 5 61 39 38 02 - Fax : +33 5 61 39 38 99
Société civile de droit français au capital de 2 000 000 € - 522 911 577 RCS Toulouse - SIRET 522 911 577 00024

marine.copernicus.eu
mercator-ocean.eu

RECORD TABLE

Issue	Date	§	Description of Change	Author	Validated By
1.0	December 2019	All	Initial version of the document	B. Buongiorno Nardelli	Mercator Ocean
2.0	May 2024	All	Used new template. Updated content to NOV24 release	B. Buongiorno Nardelli, S. Asdar	H.Etienne
3.0	May 2025	All	Used new template. Updated content to NOV25 release	B. Buongiorno Nardelli, S. Asdar, L. Della Cioppa	H.Etienne

TABLE OF CONTENTS

RECORD TABLE **2**

GLOSSARY AND ABBREVIATIONS **4**

DATA ACCESS **5**

1) INTRODUCTION **6**

a) Summary 6

2) DESCRIPTION OF THE PRODUCT SPECIFICATION **7**

b) General Information 7

c) Production System Description 8

d) Dataset details 8

e) Processing information 9

f) Catalogue updating 9

g) Time averaging 9

h) Product use and impact 10

3) FILE FORMAT **11**

4) FILES NOMENCLATURE **11**

4.1) FILE SIZE **11**

4.2) REFERENCES **12**

5) ANNEX **13**

a) Structure and semantic of NetCDF maps files for cmems_obs-mob_glo_phy-uvw_my_0.25deg-P7D-i 13

b) Structure and semantic of NetCDF maps files for cmems_obs-mob_glo_phy-uvw_nrt_0.25deg-P7D-i 14

GLOSSARY AND ABBREVIATIONS

MFC	Monitoring and Forecasting Centre
TAC	Thematic Assembly Center
ECMWF	European Centre for Medium Range Weather forecast
GLO	Global
MFWAM	Meteo France WAve Model
NRT	Near Real Time
MY	Multi Year
PUM	Product User Manual
QUID	Quality Information Document
CMEMS	Copernicus Marine Environment Monitoring Service
NetCDF	Network Common Data Form
CF	Climate Forecast (convention for NetCDF)
MDS	Marine Data Store
NRT	Near Real Time
MY	Multi Year

DATA ACCESS

After registration, you will be able to download our data. To assist you, our HelpCenter is available, and more specifically its section about download.

Information on operational issues on products and services can be found on our User Notification Service. If needed, you can contact us .

1) INTRODUCTION

a) Summary

This document describes the external product **MULTIOBS_GLO_PHY_UVW_3D_MYNRT_015_007***, which provides observation-based 3D quasi-geostrophic (QG) vertical and horizontal ocean currents from the surface to 1500 m depth over a regular grid at ¼° horizontal resolution, both in Near Real Time and as Multi Year datasets. The datasets delivered are:

- cmems_obs-mob_glo_phy-uvw_nrt_0.25deg_P7D-i, which corresponds to NRT (previous week) u, v, w weekly sampled fields (representative of each Wednesday)

- cmems_obs-mob_glo_phy-uvw_my_0.25deg_P7D-i, which corresponds to MY u, v, w weekly sampled fields (representative of each Wednesday) over the period January 1993 to year-1.

The MY dataset delivers velocities obtained by solving a Q-vector formulation of the Omega equation [DR1] applied to previous (weekly) version of ARMOR3D data (MULTIOBS_GLO_PHY_TSUV_3D_MYNRT_015_012), while the NRT dataset is based on the new daily NRT ARMOR3D dataset delivered at 1/8° resolution (downgraded to ¼° weekly resolution).

**This product supersedes former MULTIOBS_GLO_PHY_UVW_3D_REP_015_007.*

2) DESCRIPTION OF THE PRODUCT SPECIFICATION

b) General Information

Product Lines	MULTIOBS_GLO_PHY_UVW_3D_MYNRT_015_007
Geographical coverage	Global (82.125°S, 89.875°N; 179.875°W, 179.875°E), equatorial band and coastal areas are masked.
Variables	Vertical velocity Zonal and Meridional Velocity (provided as total and ageostrophic components)
Product Type	Near Real Time – Multi Year
Available time series	06/01/1993 to (day-7)
Temporal resolution	Weekly fields (analysis centered on Wednesday)
Update Frequency	Weekly
Target delivery time	weekly on Wednesday at 12 pm (noon) UTC
Horizontal resolution	1/4 degree
Number of vertical levels	75
Format	NetCDF4

c) Production System Description

Three **input data sets** are used to compute the **MULTIOBS_GLO_PHY_UVW_3D_MYNRT_015_007** product:

- the global ARMOR3D product [DR6, DR7] distributed within CMEMS as MULTIOBS_GLO_PHY_TSUV_3D_MYNRT_015_012 – old dataset-armor-3d-rep-weekly and new cmems_obs-mob_glo_phy_nrt_0.125deg_P1D-m, which provide 3D temperature (T), salinity (S), and absolute height (H);
- the multi-year L4 of global (Ekman) total velocity field at 0 m and 15 m, including zonal and meridional velocity at a 1h frequency over 1/4° regular grid (product ID: MULTIOBS_GLO_PHY_MYNRT_015_003;
- the surface fluxes provided by ECMWF (European Centre For Medium-Range Weather Forecasts) global atmospheric reanalysis ERA-5 [DR8].

The methodology used to calculate the products and **the variables** are described in the QUID documentation [DR2].

d) Dataset details

Product Lines	MULTIOBS_GLO_PHY_UVW_3D_MYNRT_015_007	
Geographical coverage	Global	
Datasets: cmems_obs-mob_glo_phy-uvw_nrt_0.25deg_P7D-i cmems_obs-mob_glo_phy-uvw_my_0.25deg_P7D-i		
Product Type	NRT	MY
Update frequency	Weekly	Weekly
Available time series	-2Y - up to real-time	1993 to year-1
Target delivery time	On wednesday at 12pm (noon) UTC	
Temporal resolution	Weekly	

Variables name in the NetCDF file and Unit: Long_name & Standard_name
uo [meter per second] quasi-geostrophic eastward velocity eastward_sea_water_velocity
vo [meter per second] quasi-geostrophic northward velocity northward_sea_water_velocity
uago [meter per second] eastward ageostrophic velocity ageostrophic_eastward_sea_water_velocity
vago [meter per second] northward ageostrophic velocity ageostrophic_northward_sea_water_velocity;
wo [meter per day] quasi-geostrophic vertical velocity upward_sea_water_velocity

e) Processing information

Product disseminating schedule :

Type of product	Product temporal coverage (hours)	Product disseminating scheduled time	Dissemination day
NRT analysis	N/A	12:00 UTC	Wednesday

f) Catalogue updating

The NRT catalogue data are not updated after the delivery of previous Wednesday analysis.

g) Time averaging

OMEGA3D data are not averaged. They are provided as daily data (representative of 00:00 UTC) sampled once per week (each Wednesday).

h) Product use and impact

This product provides observation-based 3D quasi-geostrophic (QG) ocean currents, offering unique value for research, operational oceanography, and climate monitoring by filling a critical gap in global ocean current datasets.

Key User Applications:

Ocean & Climate Research: enables studies of 3D ocean circulation, mesoscale dynamics, and vertical motions, which are essential for understanding heat and carbon transport.

Physical-biological interactions: enables studies of the interactions between 3D ocean circulation and marine life (especially phytoplankton)

Unique Value Compared to other Copernicus products:

- Observation-based: unlike model-only currents, this product derives velocities based on a combination of in situ observations and satellite data.
- 3D Vertical Velocity (w): the other observation-based Copernicus current products focus on horizontal (u, v) currents, whereas this dataset provides observational estimates of vertical motion (w).
- Long-Term Consistency (MY Product): The Multi-Year (1993–present) dataset allows climate change analyses (e.g. DR 9)
- Quasi-Geostrophic (QG) Formulation: Provides dynamically consistent velocities, making it more reliable than purely derived geostrophic currents (e.g., from altimetry alone).

3) FILE FORMAT

The products are stored using the NetCDF format.

To know more about the NetCDF format, please follow this link:
What is the format of Copernicus Marine products ? NetCDF

To understand the differences between netCDF and Zarr, please consult this article:
how-to-choose-between-netcdf-and-zarr-format-using-the-toolbox

4) FILES NOMENCLATURE

An exhaustive description of the files nomenclature when downloaded through the web portal is available here : Files Nomenclature and Format

More information on product and dataset nomenclatures is available here Products and Datasets nomenclature

4.1) FILE SIZE

DATASET NAME	NAME OF FILE	DIMENSION [GB]
cmems_obs-mob_glo_phy-uvw_my_0.25deg_P7D-i	cmems_obs-mob_glo_phy-uvw_my_0.25deg-P7D-i_YYYYMMDDThhmmZ_PYYYYMMDDThhmmZ.nc	~0.55
cmems_obs-mob_glo_phy-uvw_nrt_0.25deg-P7D_i	cmems_obs-mob_glo_phy-uvw_nrt_0.25deg-P7D-i_YYYYMMDDThhmmZ_PYYYYMMDDThhmmZ.nc	~0.55

4.2) REFERENCES

DR 1 **Buongiorno Nardelli, B.** A multi-year time series of observation-based 3D horizontal and vertical quasi-geostrophic global ocean currents. *Earth Syst. Sci. Data* 12, 1711–1723 (2020).

DR 2 **CMEMS-MOB-QUID-015-007** https://catalogue.marine.copernicus.eu/documents/QUID/CMEMS-GLO-QUID-001-024.pdf

DR 3 **Guinehut, S., A.-L. Dhomps, G. Larnicol, and P.-Y. Le Traon (2012),** High resolution 3-D temperature and salinity fields derived from in situ and satellite observations, Ocean Sci., 8(5), 845–857, doi:10.5194/os-8-845-2012.

DR4 **Mulet, S., M.-H. Rio, A. Mignot, S. Guinehut, and R. Morrow (2012),** A new estimate of the global 3D geostrophic ocean circulation based on satellite data and in-situ measurements, *Deep Sea Res. Part II Top. Stud. Oceanogr.*, *77–80*, 70–81, doi:10.1016/j.dsr2.2012.04.012.

DR5 **Asdar, S., Ciani, D. & Buongiorno Nardelli, B.** 3D reconstruction of horizontal and vertical quasi-geostrophic currents in the North Atlantic Ocean. Earth Syst. Sci. Data 16, 1029–1046 (2024).

DR6 **Guinehut, S., A.-L. Dhomps, G. Larnicol, and P.-Y. Le Traon (2012),** High resolution 3-D temperature and salinity fields derived from in situ and satellite observations, Ocean Sci., 8(5), 845–857, doi:10.5194/os-8-845-2012.

DR7 **Mulet, S., M.-H. Rio, A. Mignot, S. Guinehut, and R. Morrow (2012),** A new estimate of the global 3D geostrophic ocean circulation based on satellite data and in-situ measurements, *Deep Sea Res. Part II Top. Stud. Oceanogr.*, *77–80*, 70–81, doi:10.1016/j.dsr2.2012.04.012.

DR8 **Hersbach H, Bell B, Berrisford P, Hurihara S, Horányi A et al. (2020).** The ERA5 global reanalysis.

DR9 **Buongiorno Nardelli B., D. Iudicone (2025),** A dynamical geography of observed trends in the global ocean. *Sci. Adv*.11, eadq3532. DOI:10.1126/sciadv.adq3532

5) ANNEX

a) Structure and semantic of NetCDF maps files for cmems_obs-mob_glo_phy-uvw_my_0.25deg-P7D-i

For cmems_obs-mob_glo_phy-uvw_my_0.25deg-P7D-i_YYYYMMDDThhmmZ_PYYYYMMDDThhmmZ.nc :

```
{   dimensions:
depth = 75;
lat = 689;
lon = 1440;
    time = UNLIMITED;   // (1 currently)
variables:
    float depth(depth=75);
      :standard_name = "depth";
      :units = "m";
      :axis = "Z";
      :positive = "down";

    float lat(lat=689);
      :units = "degrees_north";
      :axis = "Y";
      :standard_name = "latitude";

    float lon(lon=1440);
      :units = "degrees_east";
      :axis = "X";
      :standard_name = "longitude";

    int time(time=1);
      :units = "days since 1981-01-01 00:00:00";
      :long_name = "time";
      :standard_name = "time";
      :calendar = "gregorian";
      :axis = "T";
      :_ChunkSizes = 1024; // int

    float wo(time=1, depth=75, lat=689, lon=1440);
      :_FillValue = -32767.0f; // float
      :long_name = "quasi-geostrophic vertical velocity";
      :standard_name = "upward_sea_water_velocity";
      :units = "m/day";
      :_ChunkSizes = 1, 15, 144, 288; // int

    float uago(time=1, depth=75, lat=689, lon=1440);
:_FillValue = -32767.0f; // float
      :long_name = "ageostrophic eastward velocity";
      :standard_name = "eastward_sea_water_velocity";
      :units = "m/s";
      :_ChunkSizes = 1, 15, 144, 288; // int
```

```
    float vago(time=1, depth=75, lat=689, lon=1440);
      :_FillValue = -32767.0f; // float
      :long_name = "ageostrophic northward velocity";
      :standard_name = "northward_sea_water_velocity";
      :units = "m/s";
      :_ChunkSizes = 1, 15, 144, 288; // int

    float uo(time=1, depth=75, lat=689, lon=1440);
      :_FillValue = -32767.0f; // float
      :long_name = "quasi-geostrophic eastward velocity";
      :standard_name = "eastward_sea_water_velocity";
      :units = "m/s";
      :_ChunkSizes = 1, 15, 144, 288; // int

    float vo(time=1, depth=75, lat=689, lon=1440);
:long_name = "quasi-geostrophic northward velocity";
      :standard_name = "northward_sea_water_velocity";
      :_FillValue = -32767.0f; // float
      :units = "m/s";
      :_ChunkSizes = 1, 15, 144, 288; // int

  // global attributes:
  :_NCProperties =
"version=1|netcdflibversion=4.4.1|hdf5libversion=1.8.17";
  :Conventions = "CF-1.7";
  :title = "Global Observed Ocean Physics 3D Quasi-Geostrophic Currents
(OMEGA3D)";
  :references = "Buongiorno Nardelli, B. A multi-year time series of
observation-based 3D horizontal and vertical quasi-geostrophic global ocean
currents. Earth Syst. Sci. Data 12, 1711–1723 (2020); Asdar, S., Ciani, D. &
Buongiorno Nardelli, B. 3D reconstruction of horizontal and vertical quasi-
geostrophic currents in the North Atlantic Ocean. Earth Syst. Sci. Data 16,
1029–1046 (2024)";
  :institution = "CNR";
  :contact = "bruno.buongiornonardelli@cnr.it";
  :netcdf_version_id = "4.9.3-development of Jun  4 2023 14:13:36 ";
  :creation_date = " "Thu Jan 18 20:32:22 2024";
  :product_version = "5.0";
  :grid_resolution = "0.25 degrees";
  :software_version = "OMEGA3D Processor V5.0";
  :methods = "Quasi-geostrophic Omega equation";
  :westernmost_longitude = "-179.875";
  :easternmost_longitude = "179.875";
  :southernmost_latitude = "-82.125";
  :northernmost_latitude = "89.875";
}
}
```

b) Structure and semantic of NetCDF maps files for cmems_obs-mob_glo_phy-uvw_nrt_0.25deg-P7D-i

For cmems_obs-mob_glo_phy-uvw_nrt_0.25deg-P7D-i_YYYYMMDDThhmmZ_PYYYYMMDDThhmmZ.nc :

```
{   dimensions:
depth = 75;
```

```
lat = 689;
lon = 1440;
    time = UNLIMITED;   // (1 currently)
variables:
    float depth(depth=75);
      :standard_name = "depth";
      :units = "m";
      :axis = "Z";
      :positive = "down";

    float lat(lat=689);
      :units = "degrees_north";
      :axis = "Y";
      :standard_name = "latitude";

    float lon(lon=1440);
      :units = "degrees_east";
      :axis = "X";
      :standard_name = "longitude";

    int time(time=1);
      :units = "days since 1981-01-01 00:00:00";
      :long_name = "time";
      :standard_name = "time";
      :calendar = "gregorian";
      :axis = "T";
      :_ChunkSizes = 1024; // int

    float wo(time=1, depth=75, lat=689, lon=1440);
      :_FillValue = -32767.0f; // float
      :long_name = "quasi-geostrophic vertical velocity";
      :standard_name = "upward_sea_water_velocity";
      :units = "m/day";
      :_ChunkSizes = 1, 15, 144, 288; // int

    float uago(time=1, depth=75, lat=689, lon=1440);
:_FillValue = -32767.0f; // float
      :long_name = "ageostrophic eastward velocity";
      :standard_name = "eastward_sea_water_velocity";
      :units = "m/s";
      :_ChunkSizes = 1, 15, 144, 288; // int

    float vago(time=1, depth=75, lat=689, lon=1440);
      :_FillValue = -32767.0f; // float
      :long_name = "ageostrophic northward velocity";
      :standard_name = "northward_sea_water_velocity";
      :units = "m/s";
      :_ChunkSizes = 1, 15, 144, 288; // int

    float uo(time=1, depth=75, lat=689, lon=1440);
      :_FillValue = -32767.0f; // float
      :long_name = "quasi-geostrophic eastward velocity";
      :standard_name = "eastward_sea_water_velocity";
      :units = "m/s";
      :_ChunkSizes = 1, 15, 144, 288; // int
```

```
    float vo(time=1, depth=75, lat=689, lon=1440);
:long_name = "quasi-geostrophic northward velocity";
      :standard_name = "northward_sea_water_velocity";
      :_FillValue = -32767.0f; // float
      :units = "m/s";
      :_ChunkSizes = 1, 15, 144, 288; // int

  // global attributes:
  :_NCProperties =
"version=1|netcdflibversion=4.4.1|hdf5libversion=1.8.17";
  :Conventions = "CF-1.7";
  :title = "Global Observed Ocean Physics 3D Quasi-Geostrophic Currents
(OMEGA3D)";
  :references = "Buongiorno Nardelli, B. A multi-year time series of
observation-based 3D horizontal and vertical quasi-geostrophic global ocean
currents. Earth Syst. Sci. Data 12, 1711–1723 (2020); Asdar, S., Ciani, D. &
Buongiorno Nardelli, B. 3D reconstruction of horizontal and vertical quasi-
geostrophic currents in the North Atlantic Ocean. Earth Syst. Sci. Data 16,
1029–1046 (2024)";
  :institution = "CNR";
  :contact = "bruno.buongiornonardelli@cnr.it";
  :netcdf_version_id = "4.9.3-development of Jun  4 2023 14:13:36 ";
  :creation_date = " "Thu May 18 20:32:22 2025";
  :product_version = "5.0";
  :grid_resolution = "0.25 degrees";
  :software_version = "OMEGA3D Processor V5.0";
  :methods = "Quasi-geostrophic Omega equation";
  :westernmost_longitude = "-179.875";
  :easternmost_longitude = "179.875";
  :southernmost_latitude = "-82.125";
  :northernmost_latitude = "89.875";
}
}
```